

2 과 · 문단 1 워크북

Science Is Not Only for Scientists (과학은 과학자만의 것이 아니다)

1. 오늘의 낱말 — 우리말 뜻을 쓰세요

1. perception

3. reserved

5. belong to

7. exclusively

2. domain

4. contributions

6. accomplishments

8. play a role in

2. 유의어·반의어 뜻 쓰기 — 우리말 뜻을 쓰세요

1. countless

3. advancement

5. remarkable

7. numerous

9. extraordinary

2. ordinary

4. few

6. progress

8. decline

3. 실전문제

1. 다음 글의 요지로 가장 적절한 것은?

In honor of the aurora chasers who first discovered the phenomenon, the name “Steve” has been kept. However, it is now written in all capital letters and stands for “Strong Thermal Emission Velocity Enhancement.” Experts have acknowledged that it is the dedicated members of the aurora chasers group that deserve the credit for discovering STEVE. If it had not been for them, it might have remained unnoticed forever. There are many interesting theories about STEVE, but none of them have been proven for certain. Scientists and citizens continue to collaborate to solve this mystery, and they are learning more and more about STEVE every day. Most of the individuals involved in these two projects are not professional scientists. Instead, they are citizen scientists — ordinary people who desire to help advance science in areas that they are interested in. Even though citizen scientists may not have science degrees or long, white lab coats, they are still capable of making valuable contributions to science. You can become one as well if you have curiosity and an interest in taking part in scientific research. The next great scientific achievement could be made by citizen scientists volunteering their time for the purpose of better understanding the world.

- ① Few people outside the expert community can contribute to scientific progress.
- ② Numerous unproven theories show that citizen science is in decline.
- ③ Ordinary people can make remarkable contributions to the advancement of science.
- ④ Extraordinary discoveries like STEVE should be left to scientists with degrees.
- ⑤ Citizen science projects are too ordinary to lead to new knowledge.

문단 1 워크북 (계속)

4. 구문 결합해서 해석하기

1. spend their days conducting experiments → _____

도움말 spend A V-ing — V-ing 하면서 A(시간)를 보내다 / their — 그들의 / days — 나날, 하루하루 / conduct — (실험 등을) 하다, 수행하다 / experiments — 실험들

2. the ability to play a role in the advancement → _____

도움말 N + to V — ~할 N / ability — 능력 / play a role in — ~에서 역할을 하다, ~에 한몫하다 / advancement — 발전, 진보

3. projects in which ordinary people have made contributions → _____

도움말 in which S' V' — S'(이/가) V'하는 / projects — 프로젝트들, 연구 과제들 / ordinary — 평범한 / people — 사람들 / have p.p. — ~해 왔다 / make — 하다 (흔한 뜻: 만들다) (made는 make의 p.p.형) / contributions — 기여, 공헌

4. There have been numerous citizen science projects → _____

도움말 have p.p. — ~해 왔다 / be — 있다 (been은 be의 p.p.형) / numerous — 수많은 / citizen science — 시민 과학 (일반 시민이 참여하는 과학 연구) / projects — 프로젝트들, 연구 과제들

5. 핵심 문장 2개 다시 해석하기

1. When we think about science, we might consider it as a domain reserved exclusively for scientists in long, white lab coats who spend their days conducting experiments and analyzing data.

나의 해석

2. There have been numerous citizen science projects in which ordinary people have made contributions to remarkable scientific accomplishments.

나의 해석

2 과 · 문단 2 워크북

Galaxy Zoo: Classifying Galaxies Together (갤럭시 주: 함께 은하를 분류하다)

1. 오늘의 낱말 — 우리말 뜻을 쓰세요

1. classify

2. suggest

3. individually

4. complain

5. tutorial

6. effectively

7. participants

8. platforms

2. 유의어·반의어 뜻 쓰기 — 우리말 뜻을 쓰세요

1. boring

2. tiny

3. brief

4. exciting

5. tremendous

6. short

7. dull

8. lengthy

9. enormous

3. 실전문제

2. 다음 글의 내용과 일치하지 않는 것은?

Kevin Schawinski was a young astronomy researcher studying black holes and the evolution of galaxies at the University of Oxford. This might sound exciting, but his job there was dull and time-consuming. He had to classify around one million images of galaxies according to their shape. Since they all looked similar but were actually shaped slightly differently, the best way to do this was to sort them while looking at each one individually. Of course, this would require a tremendous amount of time. It took Schawinski a whole week to classify just 50,000 galaxies. He thought, “But what about the remaining 950,000?” One evening after work, Schawinski met a friend named Chris Lintott and complained about the situation. Lintott suggested that he turn to the Internet to ask other people for help. Eventually, the two men ended up launching a crowdsourced online project known as Galaxy Zoo. The website was kept as simple as possible, with a very basic design and an easy-to-use interface. This allowed those who wanted to participate to get started quickly. New visitors were greeted with a brief tutorial that explained the project.

- ① Schawinski’s work of classifying galaxy images at Oxford was boring.
- ② Sorting the images one by one would require an enormous amount of time.
- ③ Lintott suggested using the Internet to get help from other people.
- ④ New visitors to the website were given a short tutorial about the project.
- ⑤ Schawinski found his task exciting because it took only a tiny amount of time.

문단 2 워크북 (계속)

4. 구문 결합해서 해석하기

1. It took Schawinski a whole week to classify → _____

도움말 It takes A B to V — A가 ~하는 데 B(시간)가 걸리다 / whole — 꼬박, 전체의 / week — 일주일 / classify — 분류하다

2. It was online media that helped spread the word → _____

도움말 It was A that V' — V'한 것은 바로 A였다 / online — 온라인 / media — 미디어, 매체 / help V — ~하는 것을 돕다 / spread the word — 소문을 퍼뜨리다

3. If it had not been for those participants → _____

도움말 If it had not been for A — A(이/가) 없었다면 / those — 그 / participants — 참여자들

4. The website was kept as simple as possible → _____

도움말 be p.p. — ~되다 / website — 웹사이트 / kept — 유지된 / as ~ as possible — 가능한 한 ~하게 / simple — 단순한

5. Galaxy Zoo has now grown into Zooniverse → _____

도움말 have p.p. — ~했다 / now — 이제 / grow into A — A로 성장하다 (grown은 grow의 p.p.형) / Zooniverse — 주니버스 (시민 과학 온라인 플랫폼 이름)

5. 핵심 문장 2개 다시 해석하기

1. It was online media that helped spread the word about Galaxy Zoo, bringing more and more participants to the website.

나의 해석

2. If it had not been for those participants, Schawinski couldn't have classified that many images.

나의 해석

2 과 · 문단 3 워크북

STEVE: A Discovery by Aurora Chasers (STEVE: 오로라 추적자들의 발견)

1. 오늘의 낱말 — 우리말 뜻을 쓰세요

1. informal

3. possess

5. acknowledged

7. deserve

2. phenomenon

4. resemble

6. dedicated

8. confirm

2. 유의어·반의어 뜻 쓰기 — 우리말 뜻을 쓰세요

1. strange

3. collaborate

5. exceptional

7. unfamiliar

9. outstanding

2. average

4. familiar

6. cooperate

8. compete

3. 실전문제

3. 다음 글의 제목으로 가장 적절한 것은?

In Alberta, Canada, there were some “aurora chasers” who formed an informal group online. They would leave their homes at night and try to get exceptional photographs of auroras. Then they would share them with the rest of the group. In the summer of 2014, the group members began to notice something strange appearing in the night sky. There were unusual ribbons of green and purple light, some of which seemed to stretch for thousands of kilometers. Sometimes they lasted for just a few minutes, while other times they remained in the sky for a whole hour. They were similar in appearance to auroras, yet they possessed some notably different features. Unsure of what they were, the group decided to name the phenomenon “Steve,” based on a scene from an animated movie where some characters give that name to an unfamiliar object. After photographing Steve for several years and discussing what it might be, the group members decided to get help from experts. They showed their photos of Steve to a professor of astronomy at a local university and a scientist at NASA. Surprisingly, neither of them had any idea what Steve was. They confirmed that it was not an aurora, but it did not resemble anything they had seen before. They realized that the group had discovered a new type of phenomenon that had never been properly studied.

- ① How Aurora Chasers Competed with NASA for the Best Photos
- ② A Strange New Light Spotted by Ordinary Sky Watchers
- ③ Why Experts Found the Lights Too Familiar to Study
- ④ An Average Aurora That Everyone Can See at Night
- ⑤ Outstanding Photos That Proved What Steve Really Was

문단 3 워크북 (계속)

4. 구문 결합해서 해석하기

1. an animated movie where some characters give that name → _____

도움말 where S' V' — S'(이/가) V'하는 (앞말을 꾸미는 관계부사) / animated — 애니메이션으로 된 / movie — 영화 / some — 몇몇의 / characters — 등장인물들 / give — 붙이다 (흔한 뜻: 주다) / that — 그 / name — 이름

2. it is the dedicated members of the aurora chasers group that deserve the credit → _____

도움말 It is A that V' — V'하는 것은 바로 A 이다 / dedicated — 헌신적인 / members — 구성원들 / of — ~의 / aurora chasers group — 오로라 추적자 단체 / deserve — ~을 받을 만하다 / credit — 공로, 인정

3. the group had discovered a new type of phenomenon → _____

도움말 had p.p. — ~했다 / group — 단체 / discover — 발견하다 (discovered 는 discover 의 p.p.형) / a new type of — 새로운 유형의 / phenomenon — 현상

4. had never been properly studied → _____

도움말 had been p.p. — ~되었다 / never — 한 번도 ~않다 / properly — 제대로 / studied — 연구된

5. Experts have acknowledged → _____

도움말 have p.p. — ~했다 / experts — 전문가들 / acknowledge — 인정하다 (acknowledged 는 acknowledge 의 p.p.형)

5. 핵심 문장 2 개 다시 해석하기

1. They realized that the group had discovered a new type of phenomenon that had never been properly studied.

나의 해석

2. Experts have acknowledged that it is the dedicated members of the aurora chasers group that deserve the credit for discovering STEVE.

나의 해석

2 과 · 문단 4 워크북

Anyone Can Become a Citizen Scientist (누구나 시민 과학자가 될 수 있다)

1. 오늘의 낱말 — 우리말 뜻을 쓰세요

1. involved in

3. desire to V

5. individuals

7. achievement

2. science degrees

4. be capable of

6. curiosity

8. volunteer

2. 유의어·반의어 뜻 쓰기 — 우리말 뜻을 쓰세요

1. expert

3. advance

5. valuable

7. professional

9. precious

2. worthless

4. amateur

6. promote

8. hinder

3. 실전문제

4. 다음 글에서 필자가 주장하는 바로 가장 적절한 것은?

When we think about science, we might consider it as a domain reserved exclusively for scientists in long, white lab coats who spend their days conducting experiments and analyzing data. They seem to have very little in common with ordinary people like us. However, this perception is far from the truth. In reality, science belongs to everyone, and we all have the ability to play a role in the advancement of science. There have been numerous citizen science projects in which ordinary people have made contributions to remarkable scientific accomplishments. Let's take a look at two of them. Kevin Schawinski was a young astronomy researcher studying black holes and the evolution of galaxies at the University of Oxford. This might sound exciting, but his job there was dull and time-consuming. He had to classify around one million images of galaxies according to their shape. Since they all looked similar but were actually shaped slightly differently, the best way to do this was to sort them while looking at each one individually. Of course, this would require a tremendous amount of time. It took Schawinski a whole week to classify just 50,000 galaxies. He thought, "But what about the remaining 950,000?"

- ① Only professional scientists should be trusted to advance science.
- ② The ideas of amateurs are worthless unless experts check them first.
- ③ Scientists in lab coats should promote their work to ordinary people.
- ④ Amateurs as well as experts can help advance science in valuable ways.
- ⑤ Citizen science projects tend to hinder real scientific progress.

문단 4 워크북 (계속)

4. 구문 결합해서 해석하기

1. areas that they are interested in → _____

도움말 N + that S' V' + 전치사 — S'(이/가) V'하는 N / areas — 분야들 / they — 그들이 / be interested in — ~에 관심이 있다

2. Even though citizen scientists may not have science degrees → _____

도움말 even though S' V' — 비록 S'(이/가) V'하지만 / citizen scientists — 시민 과학자들 / may not V — ~하지 않을 수도 있다 / have — 가지고 있다 / science degrees — 과학 학위

3. citizen scientists volunteering their time → _____

도움말 N + V-ing — ~하는 N / citizen scientists — 시민 과학자들 / volunteer — (시간·노력을) 자발적으로 내다 / their — 그들의 / time — 시간

5. 핵심 문장 2개 다시 해석하기

1. Even though citizen scientists may not have science degrees or long, white lab coats, they are still capable of making valuable contributions to science.

나의 해석

2. The next great scientific achievement could be made by citizen scientists volunteering their time for the purpose of better understanding the world.

나의 해석